

# Computer Vision

Course Introduction

2026/2  
Lecture 01



# Who am I?

- Professor in UFG at INF (2013)



Research area: **robotics and computer vision**

Some subjects:

SLAM, Visual Odometry, Stereo Vision, Camera Calibration,  
etc.

- Ph.D., Electrical Engineering, USP São Carlos (2013)
- Master, Electrical Engineering, USP São Carlos (2018)
- Computer Engineering, PUC-Go (2015)



# Schedule

- |   |                                         |    |                       |
|---|-----------------------------------------|----|-----------------------|
| 1 | Fundamentos de Processamento de Imagens | 6  | Rastreamento em Vídeo |
| 2 | Imagem como um sinal                    | 7  | Deep Learning         |
| 3 | Extração de características             | 8  | Object Detection      |
| 4 | Análise Multirresolução                 | 9  | Object Tracking       |
| 5 | Detecção de Objetos                     | 10 | Seminários            |





# Introduction

- Computer Vision course (64h)
- General Objective

To understand fundamental concepts, challenges and methods of computer vision area;

## **Critério de Avaliação**

A média final (MF) será calculada pela média ponderada das seguintes avaliações:

$$MF = \text{média}(N1, N2) * 0.5 + \text{média}(\text{Listas}) * 0.3 + \text{Seminário} * 0.2$$

onde N1 e N2 são avaliações realizadas em sala de aula. As listas serão disponibilizadas como mini-programas vinculadas aos temas abordados em sala de aula.

# Honor Code

Available at our virtual environment.

## Google Classroom

### Código de Honra da Disciplina

Este código de honra estabelece a confiança entre aluno e professor e determina as regras básicas e fundamentais para prática responsável de avaliações e trabalhos da disciplina que está sendo ministrada e contribui para uma relação harmoniosa e duradoura entre aluno e professor.

Segue o código:

**O Código de Honra é um compromisso dos alunos, individual e coletivamente:**

- que não darão nem receberão auxílio nos exames;
- que não darão ou receberão ajuda não permitida em trabalhos de classe, na preparação de relatórios ou em qualquer outro trabalho que deva ser usado pelo instrutor como base de classificação;
- que eles farão sua parte e participarão ativamente de fazer com que outros, assim como eles próprios, mantenham o espírito e a letra do Código de Honra.
- O professor e a Unidade Acadêmica, por sua vez, manifesta sua confiança na honra de seus alunos, abstendo-se de supervisionar exames e de tomar precauções incomuns e desarrazoadas para evitar as formas de desonestidade mencionadas acima.
- O corpo docente também evitará, na medida do possível, procedimentos acadêmicos que criem tentações para violar o Código de Honra.
- Embora o corpo docente sozinho tenha o direito e a obrigação de definir os requisitos acadêmicos, os alunos e o corpo docente trabalharão juntos para estabelecer as condições ideais para um trabalho acadêmico honroso.

Atenciosamente,

**Prof. Gustavo T. Laureano.**



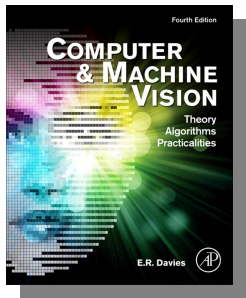
# Google Classroom Code



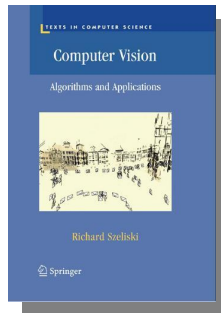
zjgrbhsq

# Introduction

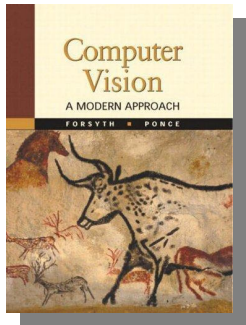
Many of the words of this course are taken from the books:



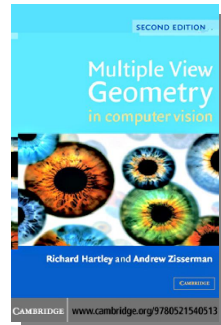
*Computer & Machine Vision – Theory, Algorithms, Practicalities.*  
Fourth Edition, 2012. Elsevier Inc.  
E. Roy Davies



*Computer Vision – Algorithm and Applications.*  
2010. Springer.  
Richard Szeliski



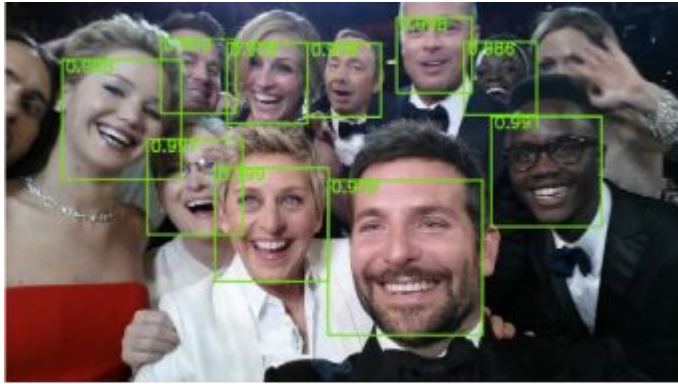
*Computer Vision – A modern Approach.*  
2003. Prentice Hall.  
David Forsyth and Jean Ponce



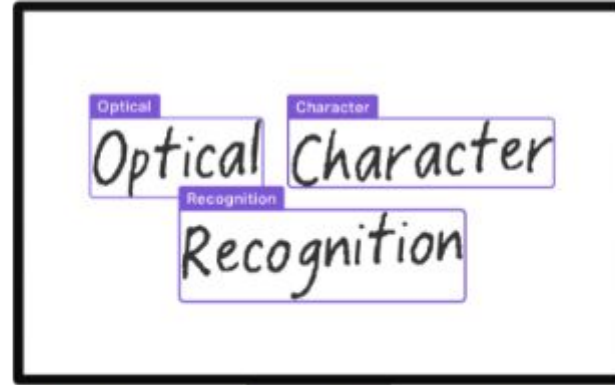
*Multiple View Geometry.*  
Second Edition, 2012. Cambridge.  
Richard Hartley & Andrew Zisserman



# What is Computer Vision?



Face Detection



Optical Character Recognition (OCR)

# What is Computer Vision?

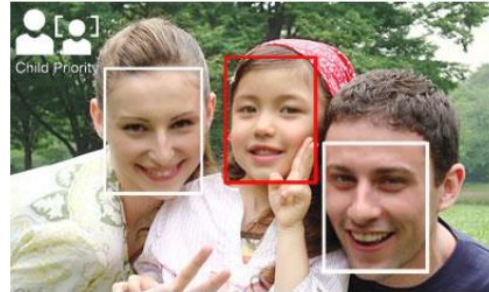


Automated Industrial production line optical inspection

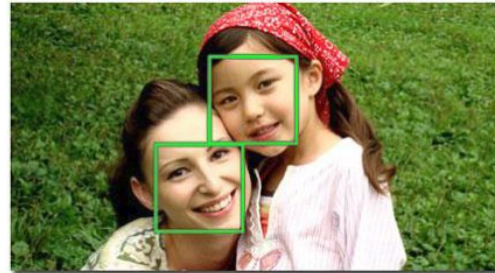
# What is Computer Vision?



Sony Cyber-shot



Age recognition



Smile recognition

Face detection

# What is Computer Vision?



Plant identification



# What is Computer Vision?



Smart glasses



# What is Computer Vision?



Offside line analysis





# What is Computer Vision?



Image Classification



Panorama  
Stitching

# What is Computer Vision?

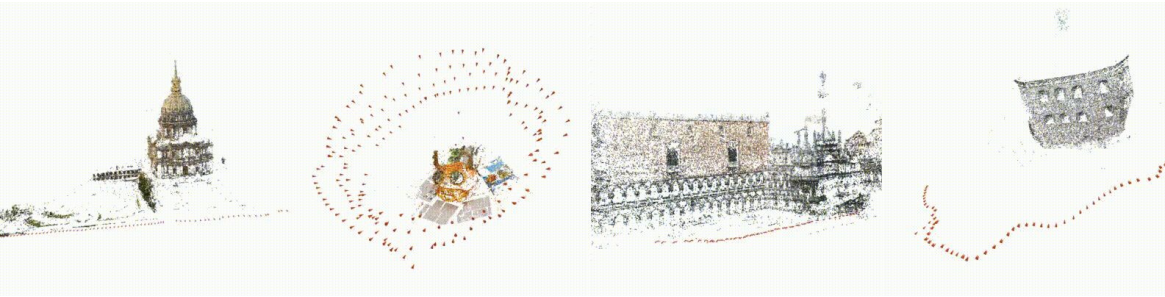


Surveillance

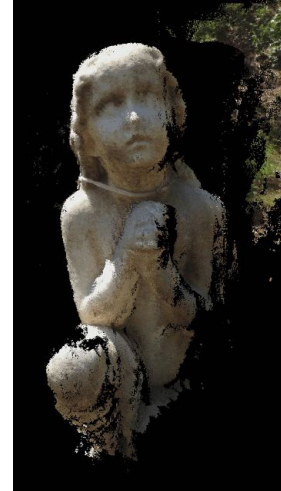


Semantic Segmentation

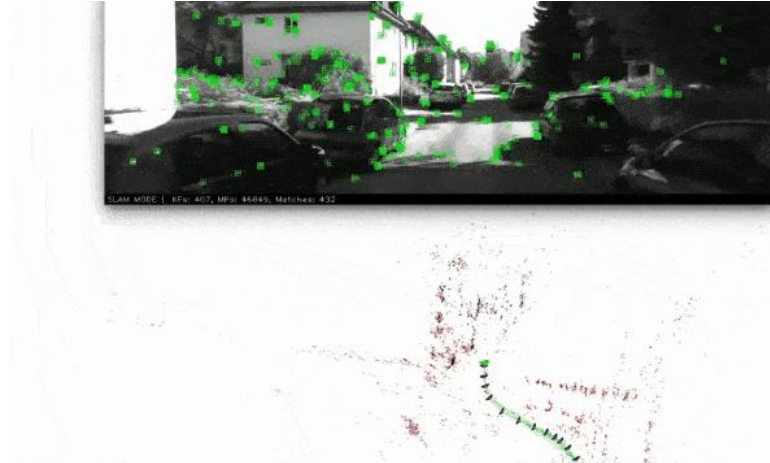
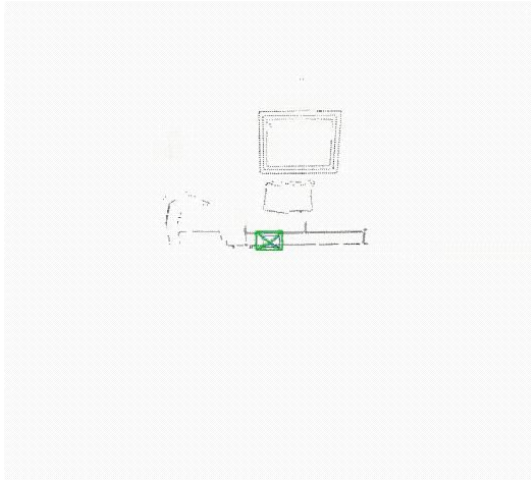
# What is Computer Vision?



Structure from Motion



# What is Computer Vision?

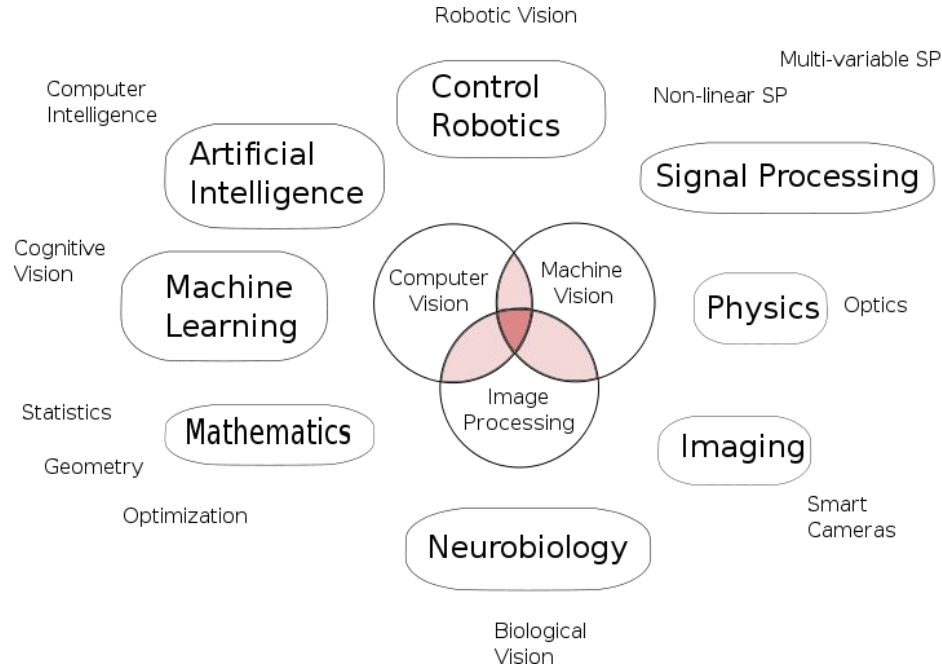


Visual Odometry and Simultaneous Localization and Mapping



**Computer  
vision is much  
more beyond  
image  
classification  
and processing**

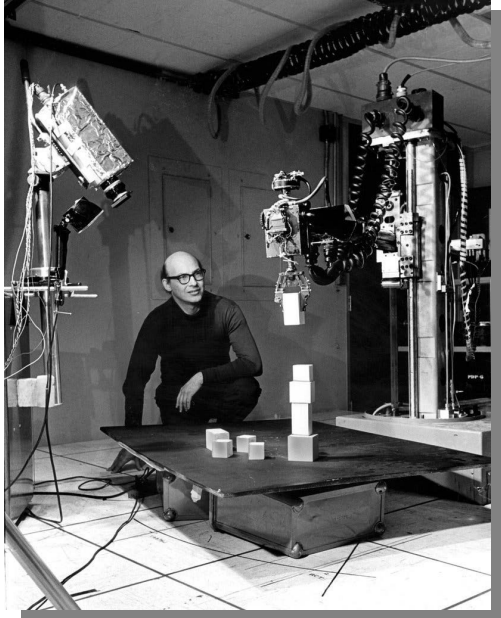
# What is Computer Vision?



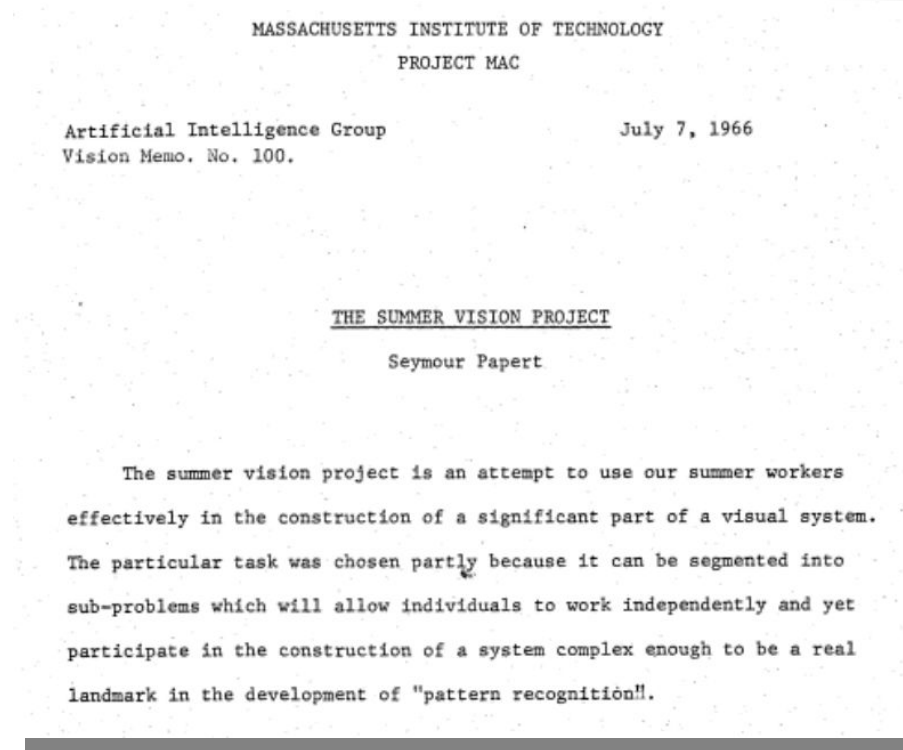
Relation between computer vision and various other fields.  
[https://en.wikipedia.org/wiki/Computer\\_vision](https://en.wikipedia.org/wiki/Computer_vision)



# The Origin of Computer Vision



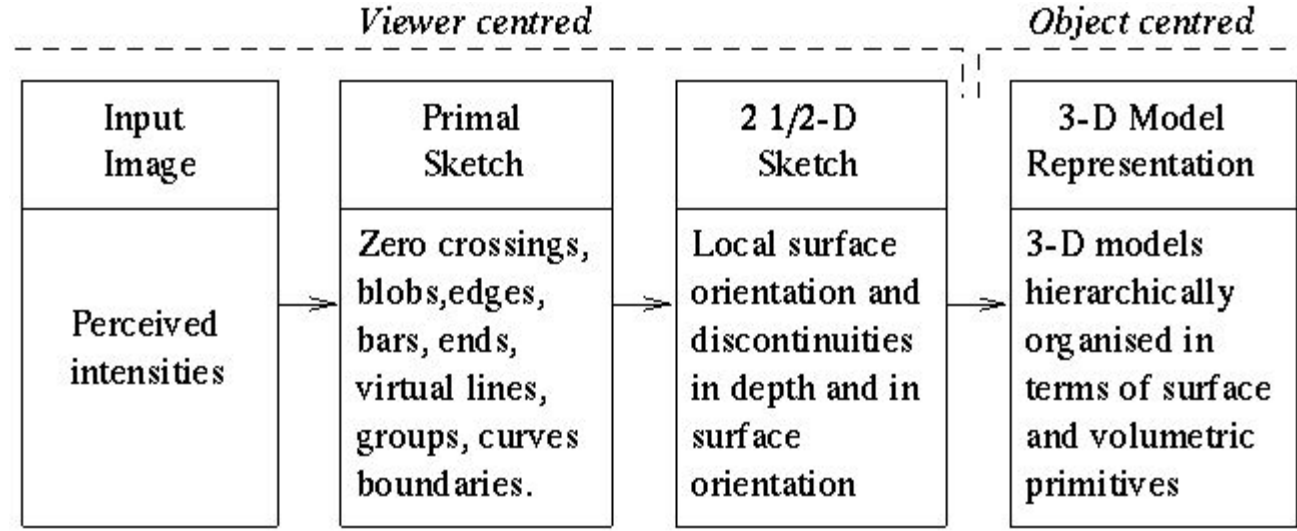
Marvin Minsky in a lab at MIT  
in 1968





# Primal Model of Computer Vision System

David Marr's Theory of Vision (Neuroscientist)



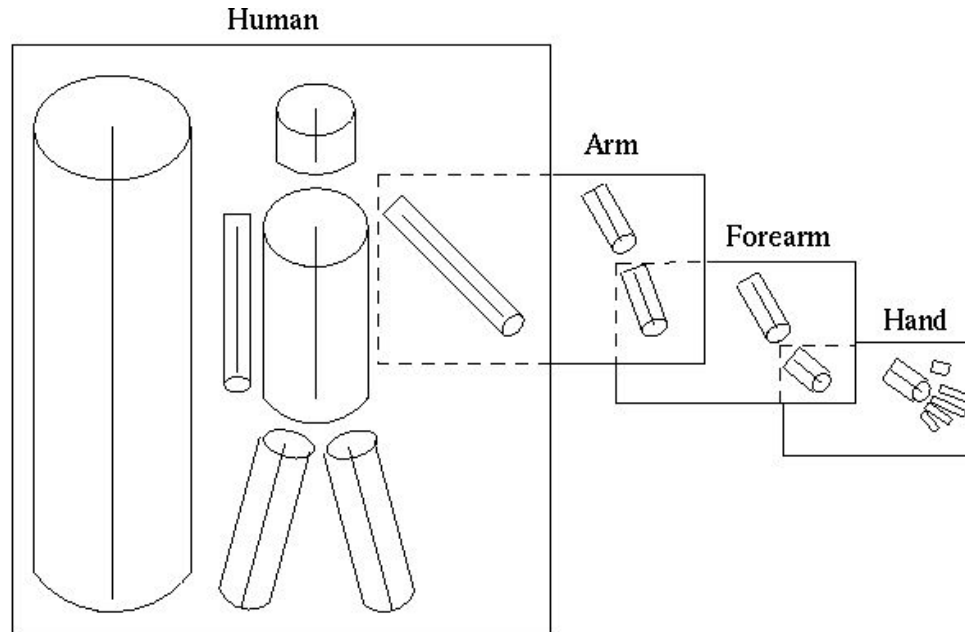
[https://homepages.inf.ed.ac.uk/rbf/CVonline/LOCAL\\_C](https://homepages.inf.ed.ac.uk/rbf/CVonline/LOCAL_C)

D. Marr. Vision. W. H. Freeman and Co., 1982.



# Primal Model of Computer Vision System

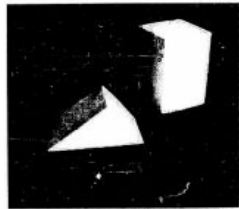
David Marr's Theory of Vision (Neuroscientist)



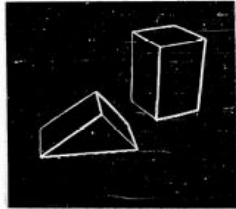
An example of Marr's 3-D model description (taken from [\[Mar82\]](#)). Each box corresponds to a 3-D model. The left side of a box contains the model axis and the right side contains the decomposition of the model's component axes.

# A Brief History

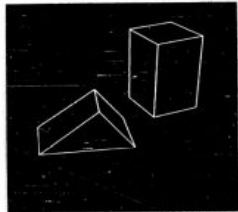
- 1970s



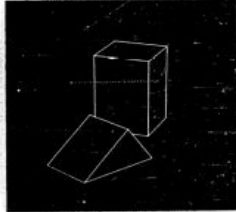
A. Original Picture



B. Differentiated Picture

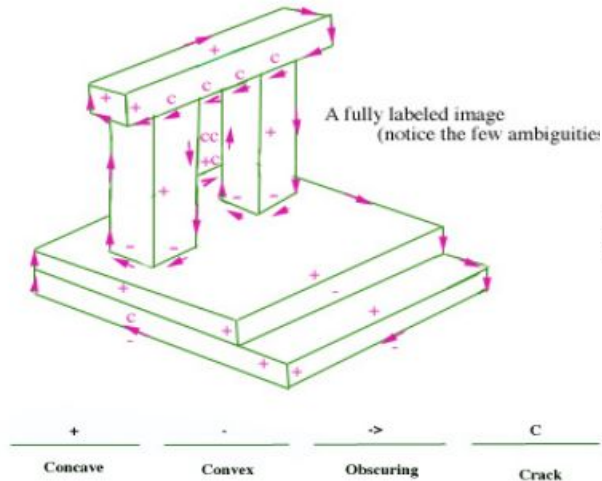


C. Line Drawing

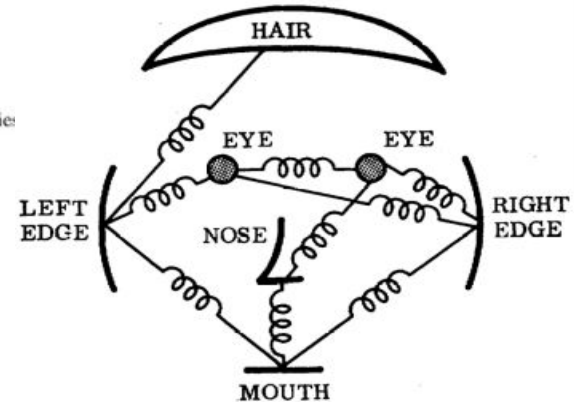


D. Rotated View

Roberts: Machine perception of three-dimensional solids. PhD Thesis, 1963



Line Labeling



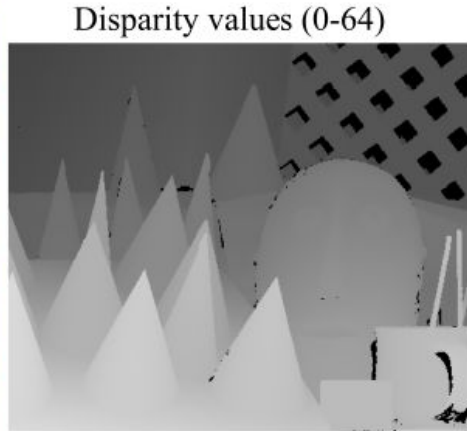
Pictorial Structure  
Fischler and Elschlager 1973

# A Brief History

- 1980s



Stereo  
Correspondence



Note how disparity is larger  
(brighter) for closer surfaces.

ow algorithm



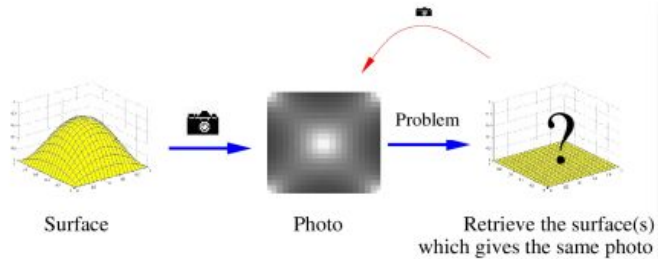
Optical  
Flow



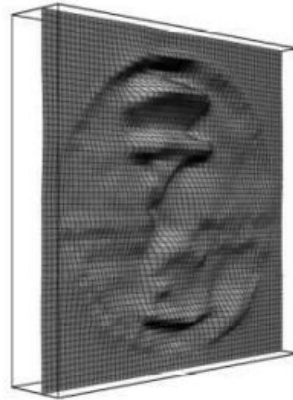
# A Brief History

- 1980s

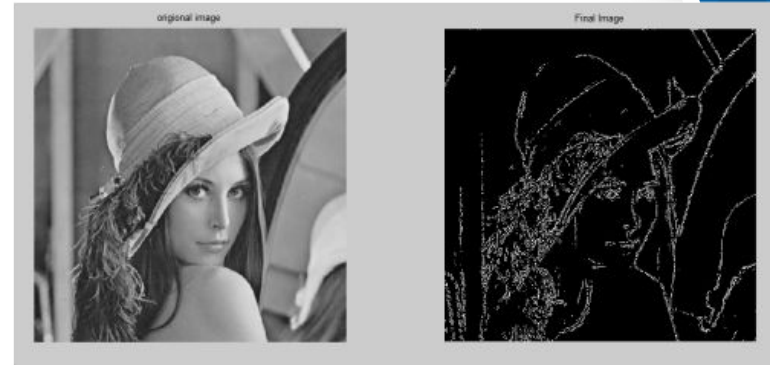
Shape from X techniques (shape from shading, shape from texture, shape from shadows) Edge and color



Shape from shading



Freeman and Adelson  
1991



Canny edge detector. Canny, 1986

# A Brief History

- 1980s

Markov Random Fields (MRFs)

$$E(x) = \sum_i \underbrace{\Psi_i(x_i)}_{\text{Unary}} + \sum_{i \sim j} \underbrace{\Psi_{i,j}(x_i, x_j)}_{\text{Pairwise}}$$

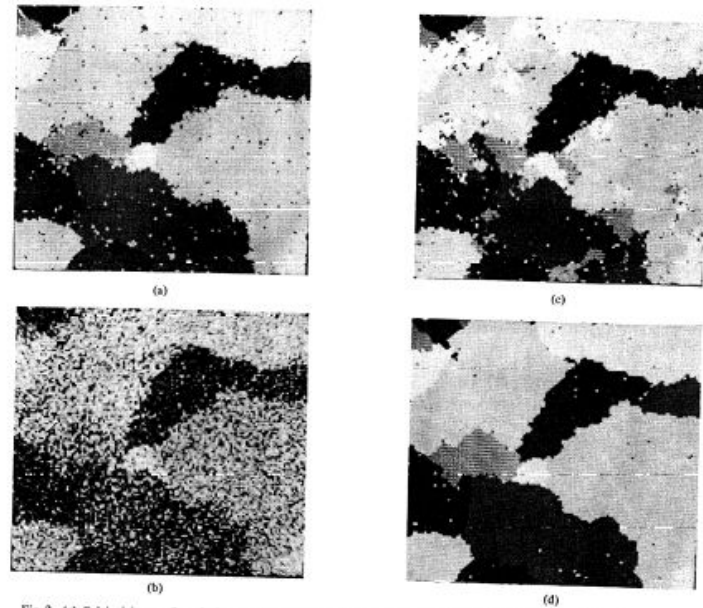


Fig. 2. (a) Original image: Sample from MRF. (b) Degraded image: Additive noise. (c) Restoration: 25 iterations. (d) Restoration: 300 iterations.

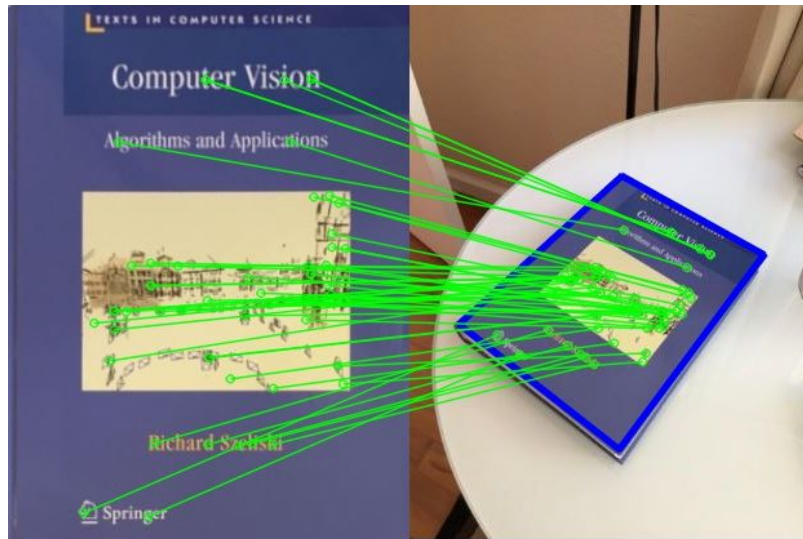
Geman and Geman: Stochastic Relaxation, Gibbs Distributions, and the Bayesian Restoration of Images. PAMI, 1984



# A Brief History

- 1990s

## Structure from Motion and Multi-view Reconstruction



David Lowe: Object recognition from local scale-invariant features. ICCV, 1999.





# A Brief History

- 1990s



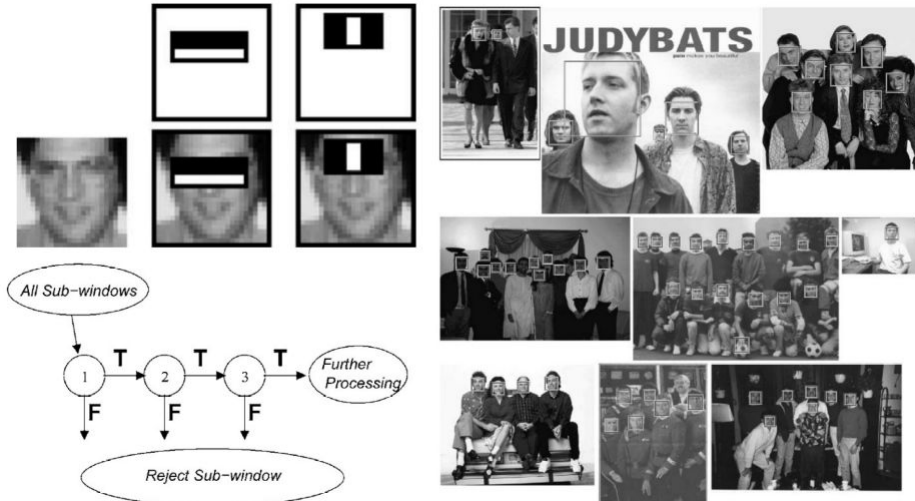
ring

Turk and Pentland: Face  
recognition using Eigenfaces.  
CVPR, 1991

# A Brief History

- 2000s

Data-driven and learning approaches



Viola and Jones: Robust Real-time Object Detection. IJCV, 2001.

# A Brief History

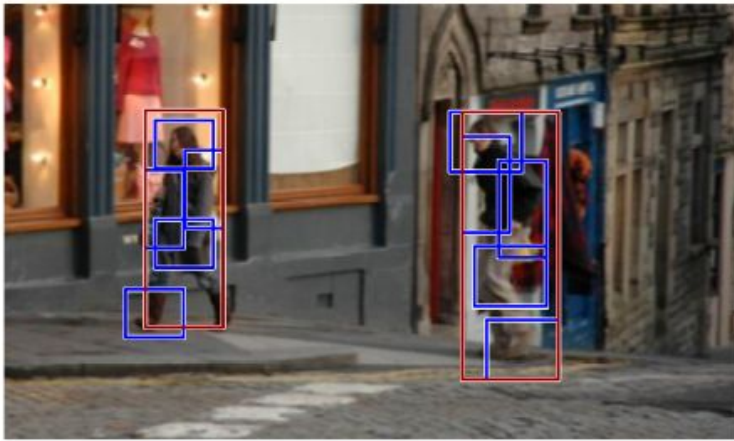
- 2000s



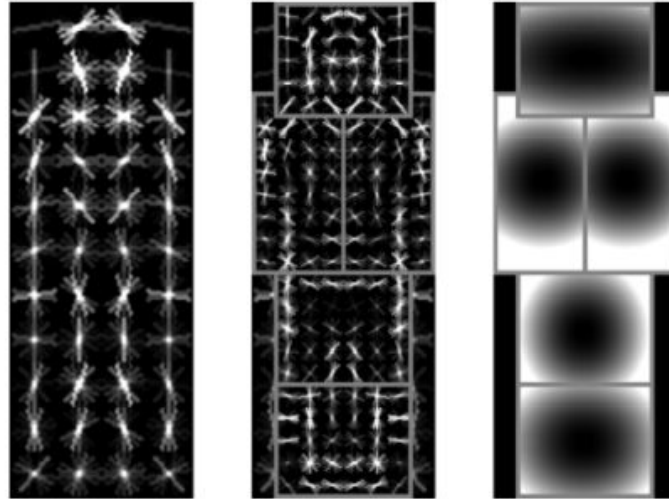
Dalal and Triggs: Histograms of Oriented Gradients for Human Detection. CVPR, 2005.

# A Brief History

- 2000s



c



# A Brief History

- 2000s

Datasets age



**The PASCAL Visual Object Classes Challenge 2007**



PASCAL VOC, Everingham et al., 2005 - 2012



ImageNet, Deng et al., 2009



# A Brief History

- 2000s

## Large scale Structure From Motion (SFM)



San Marco Square: 13,699 images, 4,515,157 points

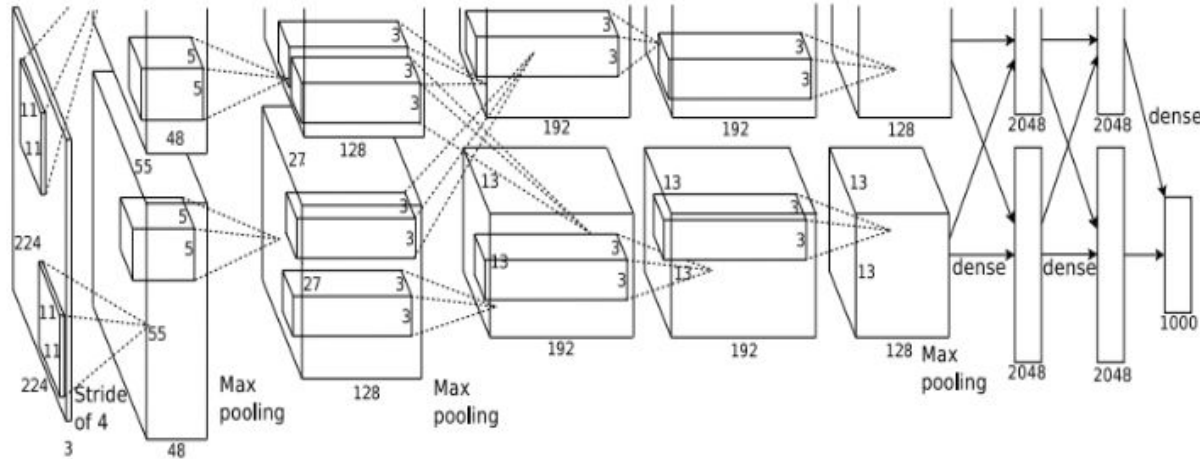
Agarwal et al. Building Rome in day. ICCV, 2009.





# A Brief History

- 2010s

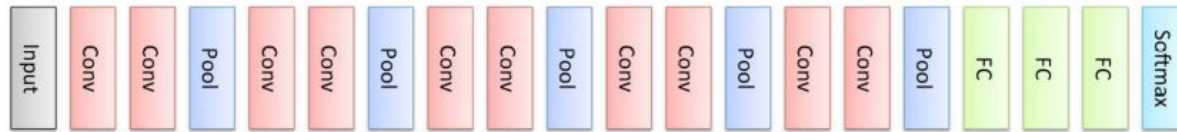


AlexNet. Krizhevsky et al., 2012, designed for ImageNet classification

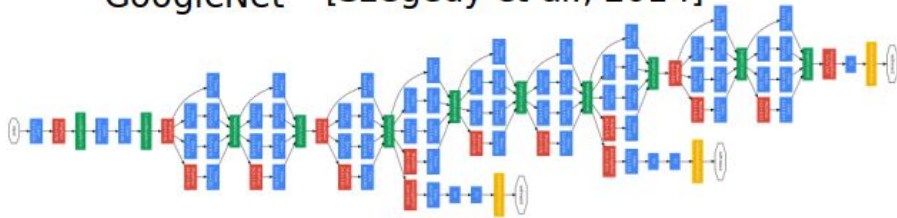
# A Brief History

- 2010s

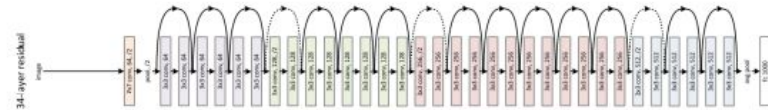
Deeper and wider networks  
VGGNet [Simonyan and Zisserman, 2014]



GoogleNet [Szegedy et al., 2014]

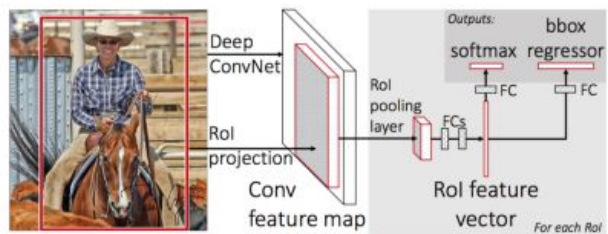


ResNet [He et al., 2015]

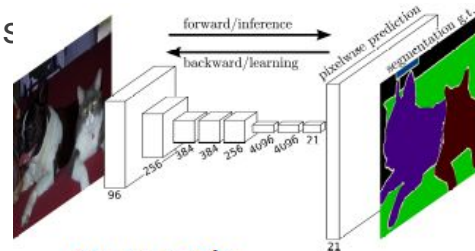


# A Brief History

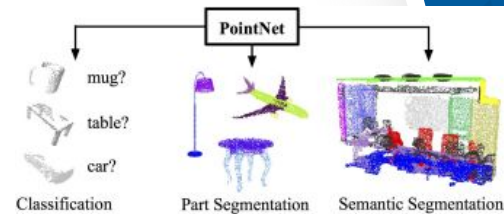
- 2010s



on and s



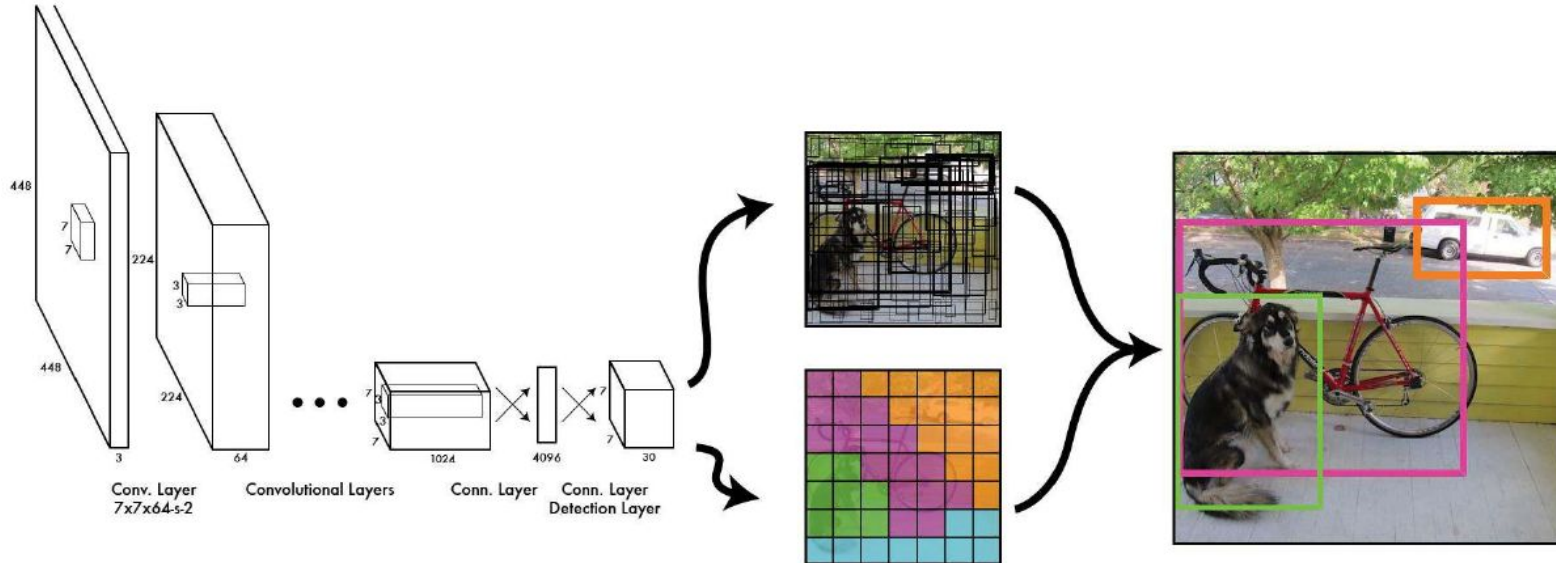
Semantic Segmentation (FCN, Long et al., 2014)



Point Cloud Recognition (PointNet, Qi et al., 2016)

# A Brief History

- 2010s



You Only Look Once (Object Detection) YOLO v1



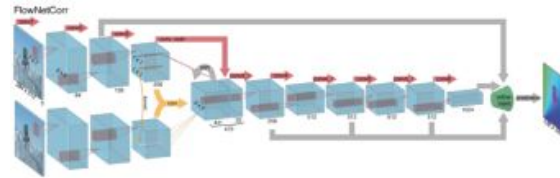
# A Brief History

- 2010s

Deep Learning for optical flow and depth estimation



Depth Estimation (Eigen et al. 2014)



Optical Flow  
(FlowNet Fischer et al.  
2015)



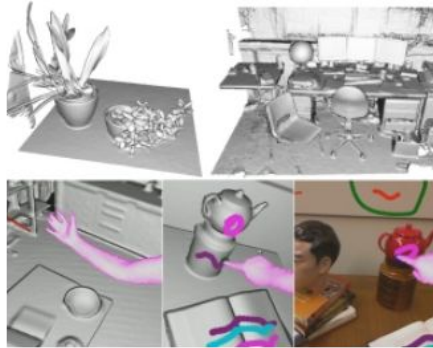
# A Brief History

- 2010s

Depth sensing and 3D Vision



Microsoft Kinect, 2010



KinectFusion, Newcombe et al., 2011

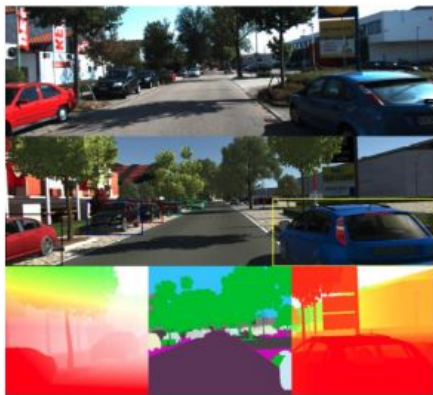


DynamicFusion, Newcombe et al., 2015

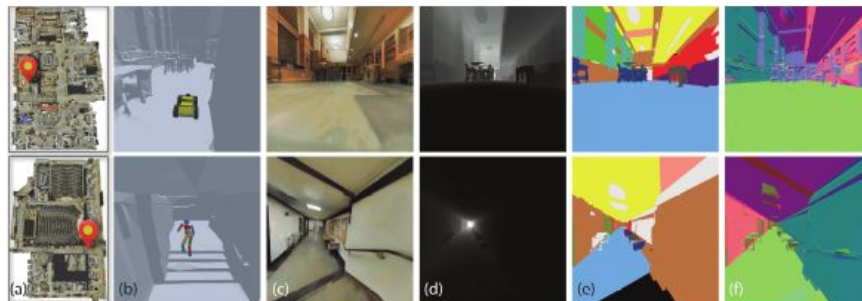
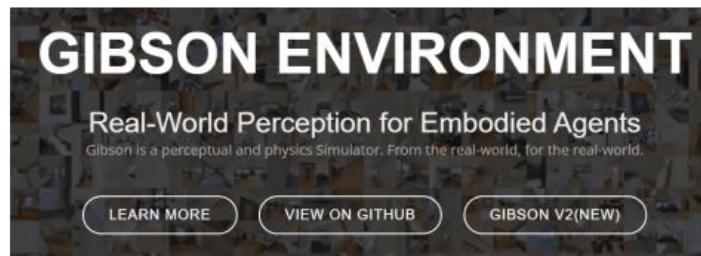


# A Brief History

- 2010s



The KITTI dataset, Geiger et al., 2012

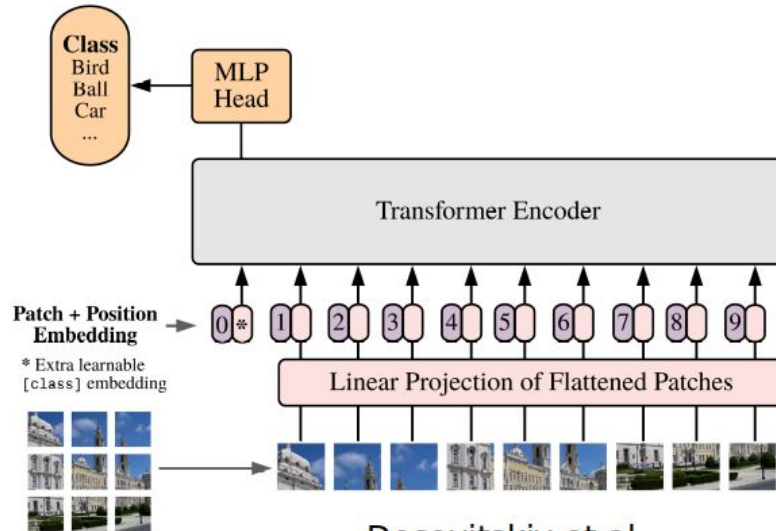


The Gibson environment, Xia et al., 2018



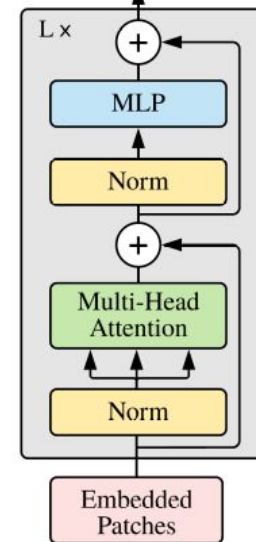
# A Brief History

- 2020s



Dosovitskiy et al.,  
ICLR'21

Transformer Encoder





# Why Computer Vision is Hard?

- Computer Vision, the challenge.

vision is the sense that man has come to depend upon above all others;

the eyes provide megabits of information that exceed 10 megabits per second (mbit/s);

much of this information is redundant and filtered by the cortex layers;

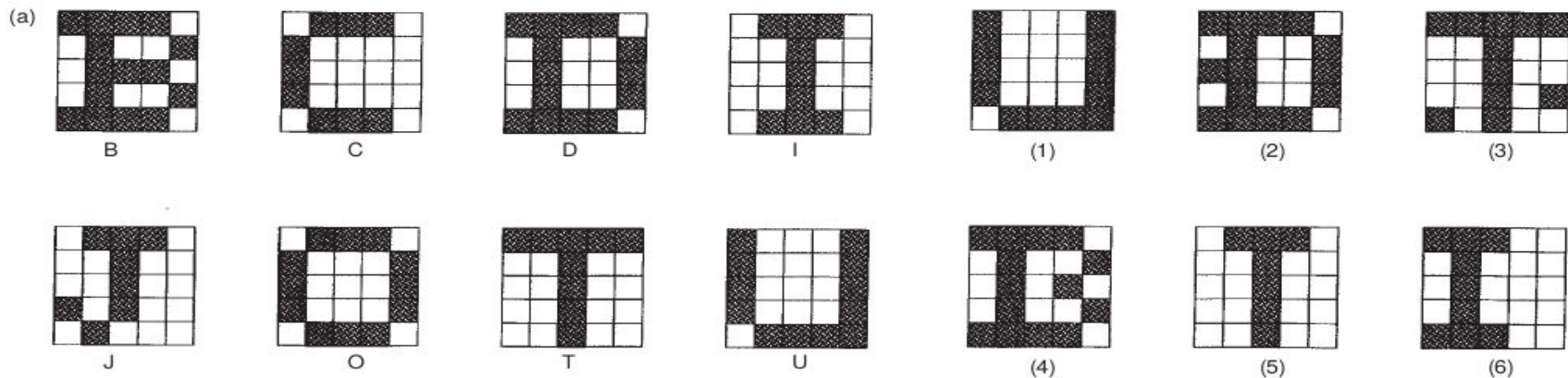
the amount of information the higher centers receive from the eyes must be at least **two orders of magnitude greater than** all the information they obtain from the other senses.



# Why Computer Vision is Hard?

## Typical Problems of Computer Vision: **Recognition**

(b)



Some simple 25-bit patterns and their recognition classes used to illustrate some of the basic problems of recognition: (a) training set patterns (for which the known classes are indicated); (b) test patterns.

(E. R. Davies, 2012)

# Why Computer Vision is Hard?

Typical Problems of Computer Vision: **Recognition**



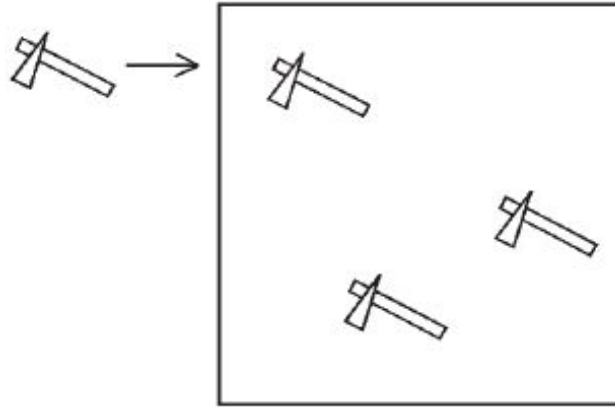
Image appearance changes dramatically over time due to non-rigidity of body and clothing and lighting.





# Why Computer Vision is Hard?

Typical Problems of Computer Vision: **Object location**



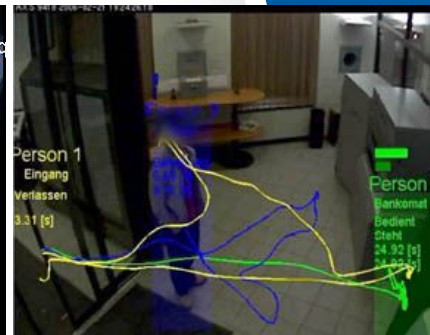
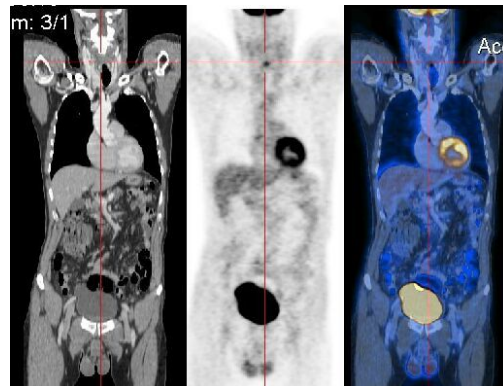
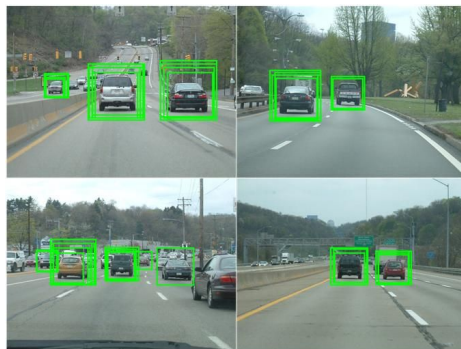
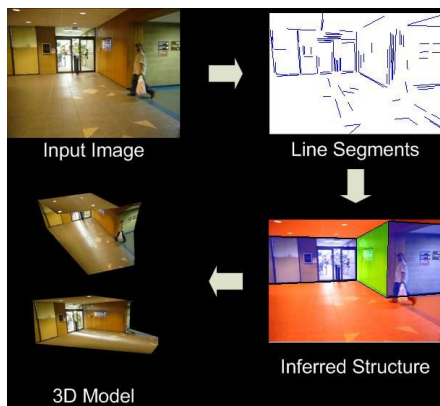
Template matching, the process of moving a suitable template over an image to determine the precise positions at which a match occurs—hence revealing the presence of objects of a particular type.

(E. R. Davies, 2012)



# Why Computer Vision is Hard?

## Typical Problems of Computer Vision: Scene Analysis



Examples of scene analysis applications. Left to right: 3D scene analysis (reconstruction), object tracking, medical images and motion analysis.

<http://www.caa.tuwien.ac.at/cvl/research-areas/video-analysis/>  
<http://www.cs.cmu.edu/~dclee/>  
<http://www.cs.cmu.edu/~hebert/scene.htm>

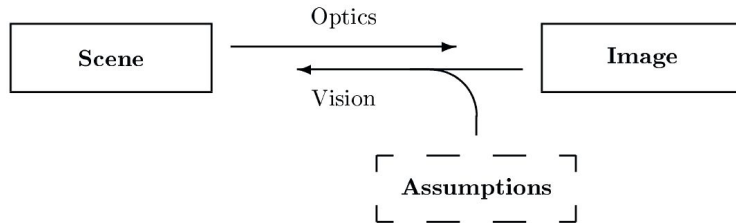


# Why Computer Vision is Hard?

## Vision as Inverse Optics/Vision as Inverse Graphics

**Graphics** is a “feedforward” activity, i.e. images can be produced straightforwardly once sufficient specification about the viewpoint and the objects, and knowledge of the laws of image formation, has been obtained. (require a lot of computation – **deterministic task**)

The situation is not so straightforward for **vision** because search is involved and there is an accompanying combinatorial explosion. (require a lot of computation – **no-deterministic task**)



Vision as inverse optics. (H. A. Mallot, 2000)



# Information in images

Light source



# Information in images

Physical interactions of a ray of light with the visible world attach certain information about the world to the ray

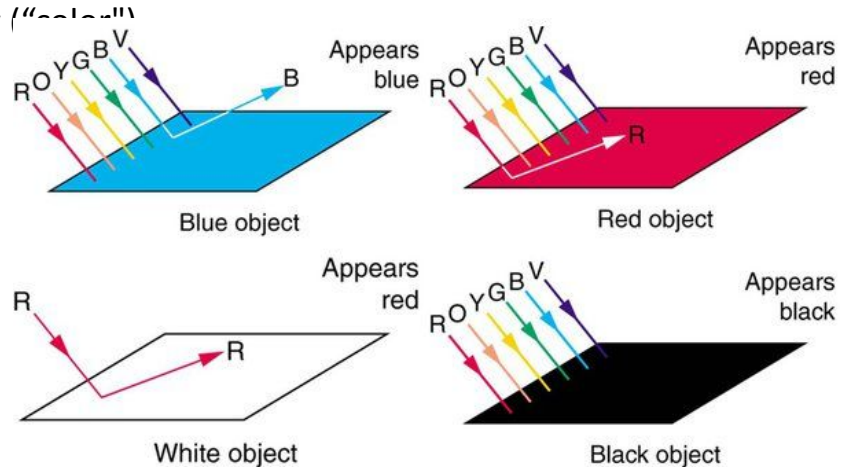
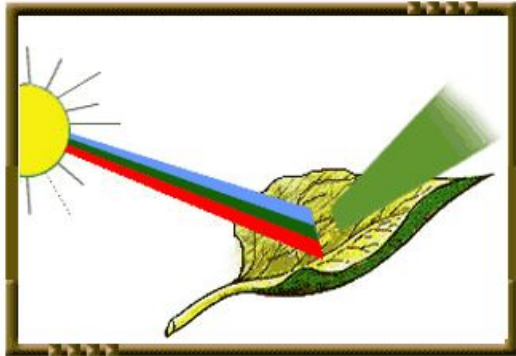


# Information in images

Physical interactions of a ray of light with the visible world attach certain information about the world to the ray

- **Reflection** (Opaque surfaces):

surface properties such as reflectivity ("color")





# Information in images

Physical interactions of a ray of light with the visible world attach certain information about the world to the ray

- **Reflection** (Opaque surfaces):

surface properties such as reflectivity ("color"),  
albedo ("brightness")



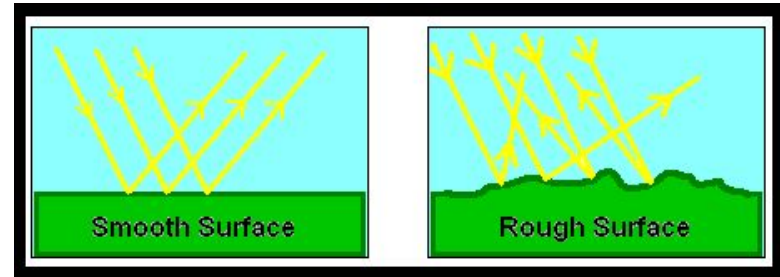
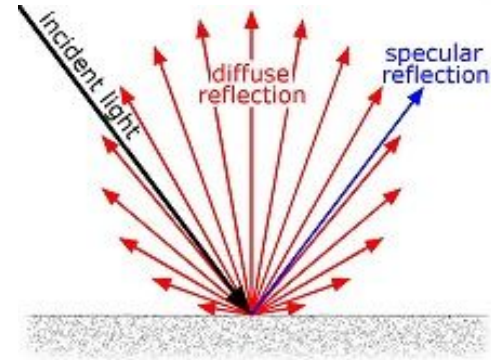
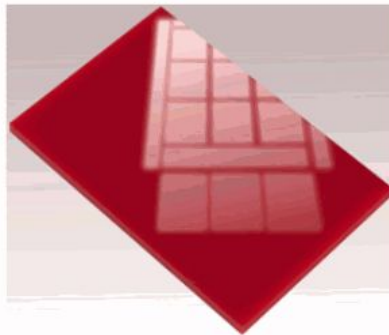
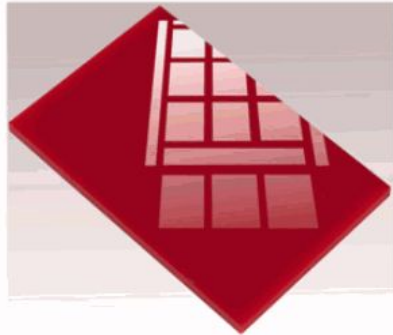


# Information in images

Physical interactions of a ray of light with the visible world attach certain information about the world to the ray

- **Reflection** (Opaque surfaces):

surface properties such as reflectivity ("color"),  
albedo ("brightness")  
smoothness ("glossy" or "dull")



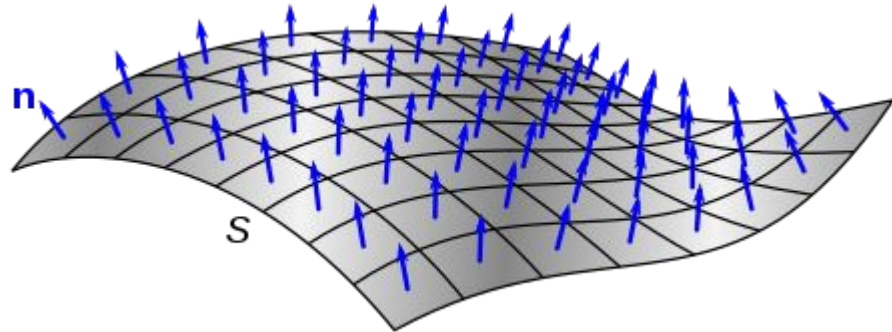


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- **Reflection** (Opaque surfaces):

surface properties such as reflectivity ("color"),  
albedo ("brightness")  
smoothness ("glossy" or "dull")  
orientation of the surface.





# Information in images

Physical interactions of a ray of light with the visible world attach certain information about the world to the ray

- **Scattering** (transparency): Scattering can reveal physical characteristics of the medium or the position (distance) of the light source.





# Information in images

Physical interactions of a ray of light with the visible world attach certain information about the world to the ray

- **Refraction:** Refraction materials and light source. Clues about physical properties of the material.

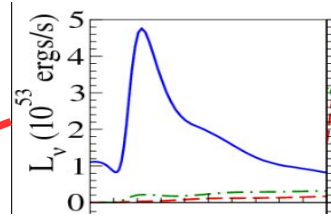
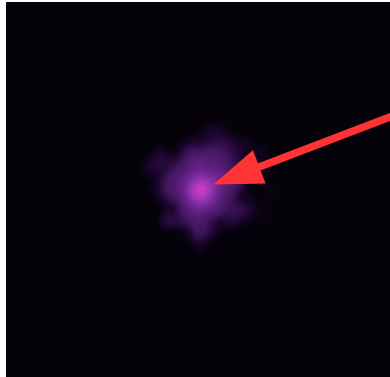




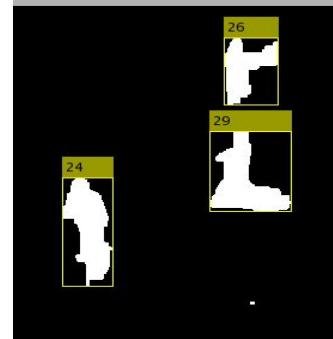
# Information in images

Information in the distribution of incident rays of light

- Temporal



Star monitoring

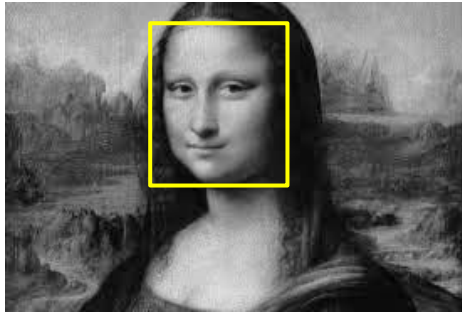


Pixel changes detection

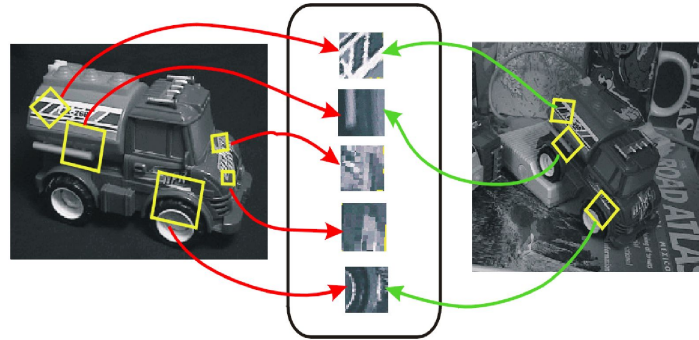
# Information in images

Information in the distribution of incident rays of light

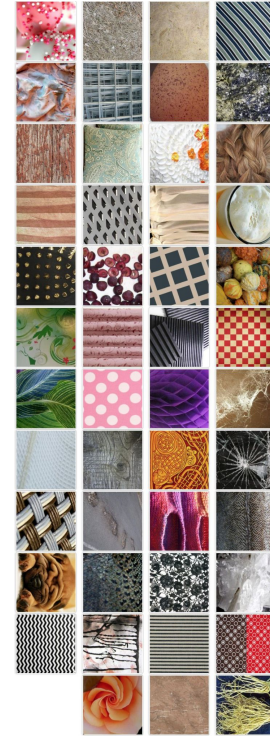
- Spatial



Feature detection



Feature correspondence



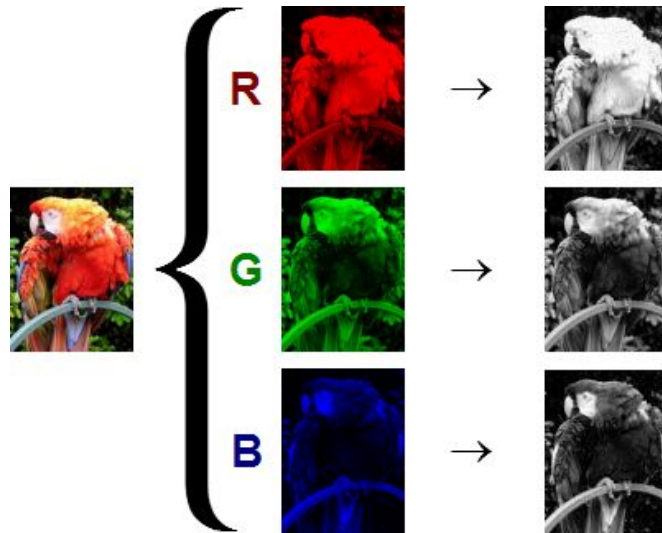
Texture



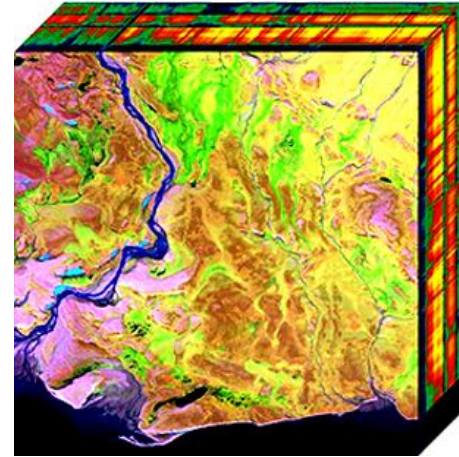
# Information in images

Information in the distribution of incident rays of light

- Spatial (Multiple channels)



RGB channels



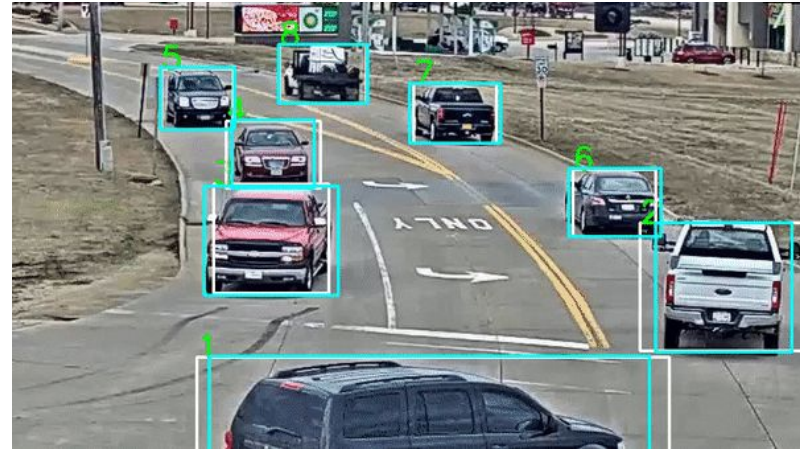
Hyperspectral Images



# Information in images

Information in the distribution of incident rays of light

- Spatio-Temporal (video)

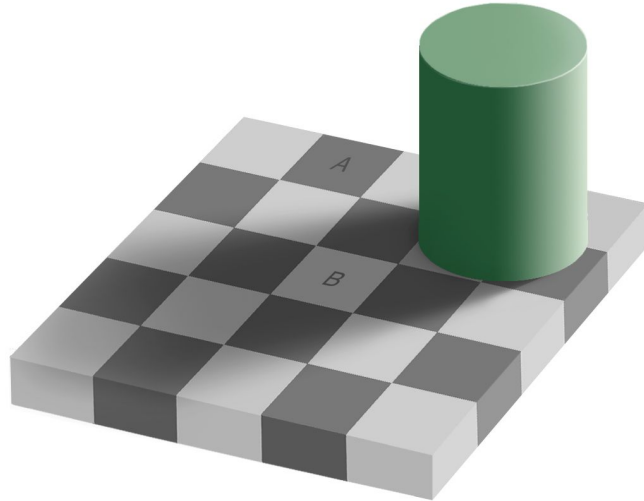




# Information derived from images

## Surface colors

Perceiving surfaces unchanged under a wide range of different illumination is an example of a perceptual constancy, a very common but often unnoticed phenomenon of perception.



Example of color constancy phenomena.

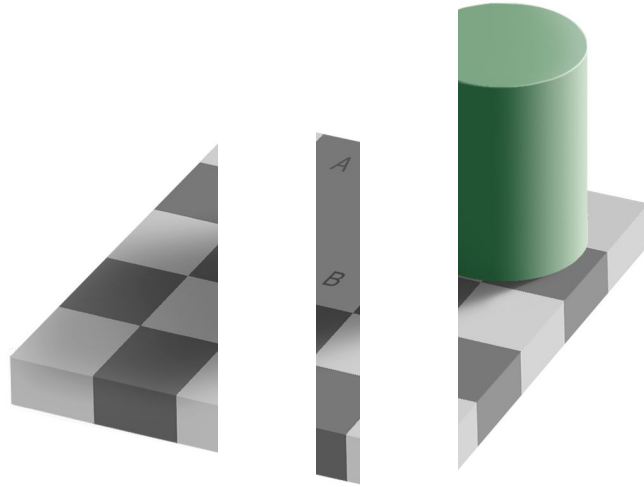
([https://en.wikipedia.org/wiki/Color\\_constancy](https://en.wikipedia.org/wiki/Color_constancy))



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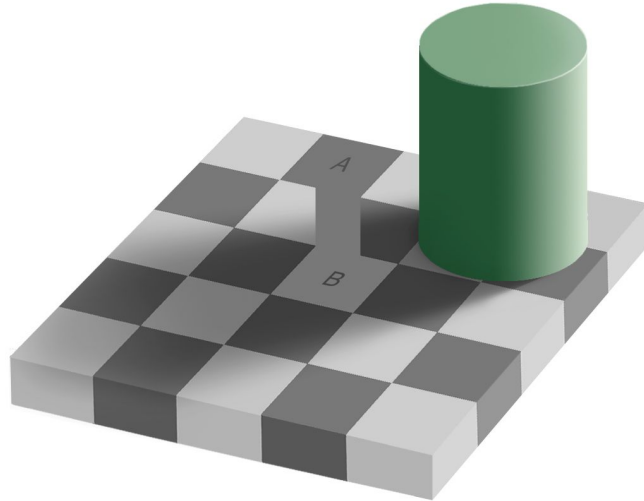
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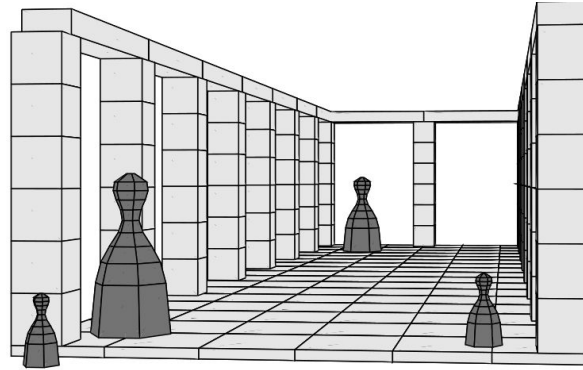
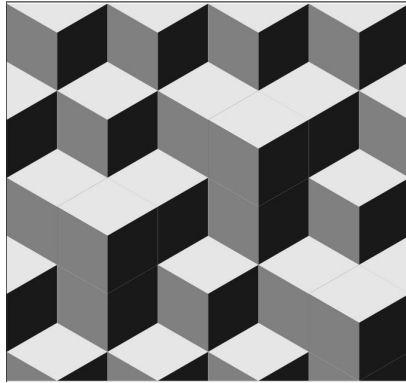
([https://en.wikipedia.org/wiki/Color\\_constancy](https://en.wikipedia.org/wiki/Color_constancy))



# Information derived from images

## Depth, size and subtended angle

To discern the distance of individual points from the observer, as well as the shape of surfaces and objects from their images.

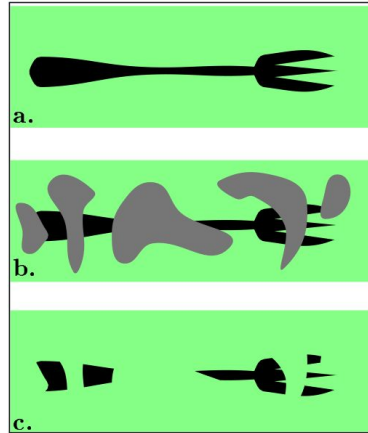


Example of depth information from shading and geometry, sizes and subtended angle.

(H. A. Mallot, 2000)

# Information derived from images

## Perceptual organization



Example of perceptual organization. Left to right, subjective contours, perceptual grouping with occlusions, segmentation.

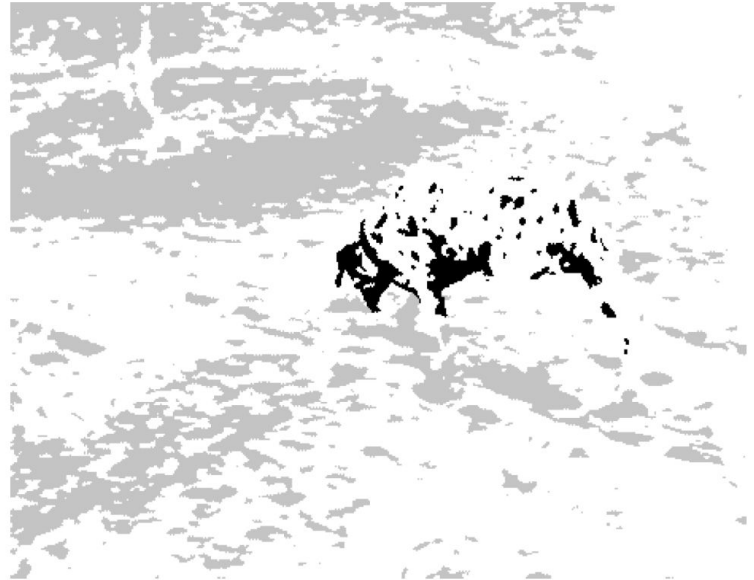
(H. A. Mallot, 2000)





# Information derived from images

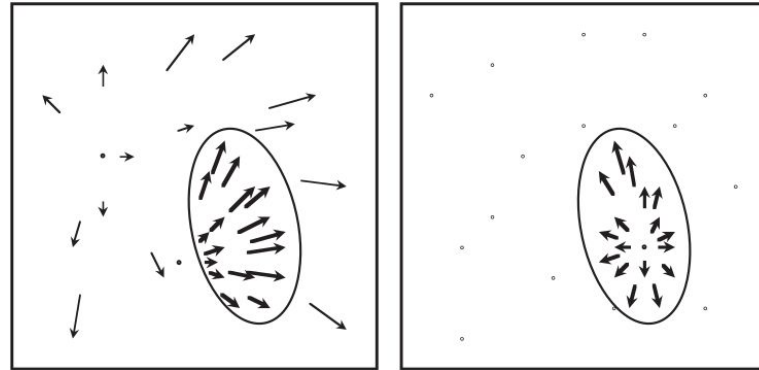
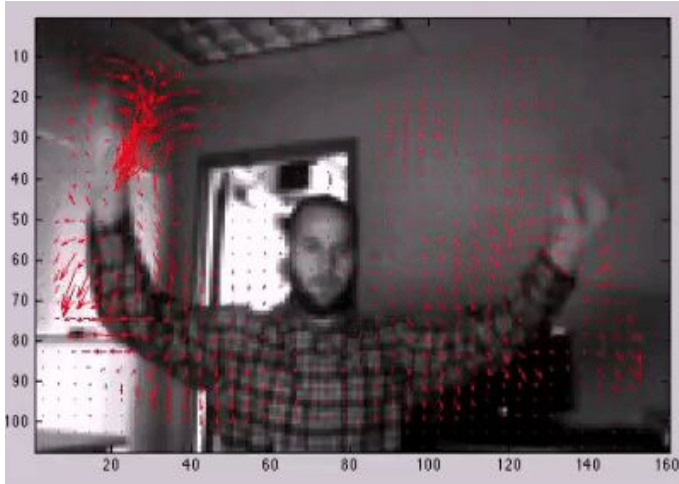
## Perceptual organization





# Information derived from images

## Motion



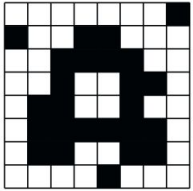
Examples of motion in images. Interpretation of velocity flow fields.

(E. R. Davies, 2012)

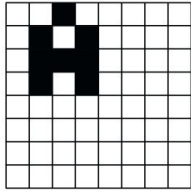


# Information derived from images

## Object, functionality and pattern recognition



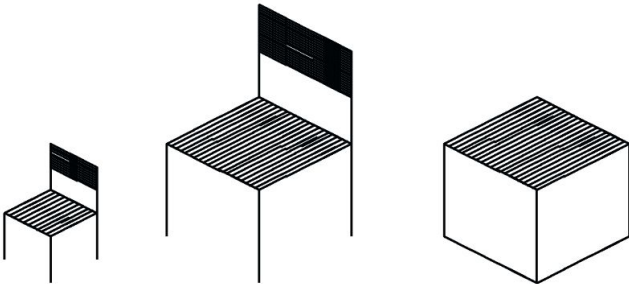
a. Noise



b. Invariance

A A  
TAE CAT

c. Context



d. Functionality

Examples of types of pattern recognition problems. Visual patterns and functionality.

(H. A. Mallot, 2000)



# Information derived from images

## Shapes





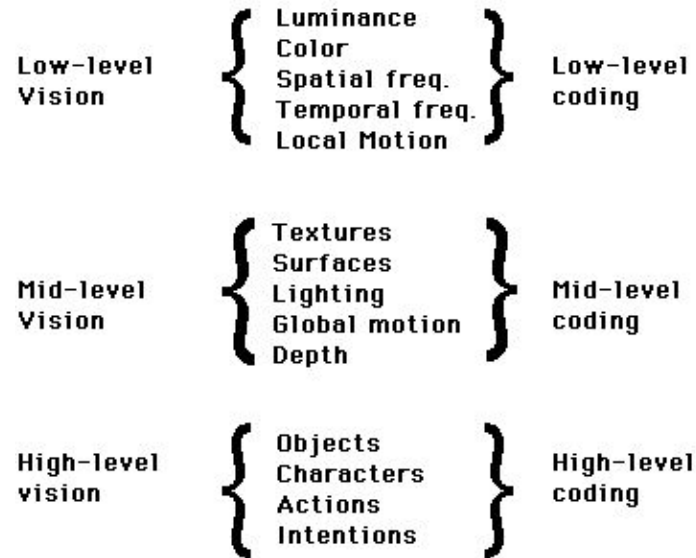


# Levels of Complexity

Low-Level Vision (Early Vision)

Intermediate-level Vision

High-Level Vision



Levels of complexity in Computer Vision.

(<http://xenia.media.mit.edu/~dfr/apw-summary/apw-summary.html>)

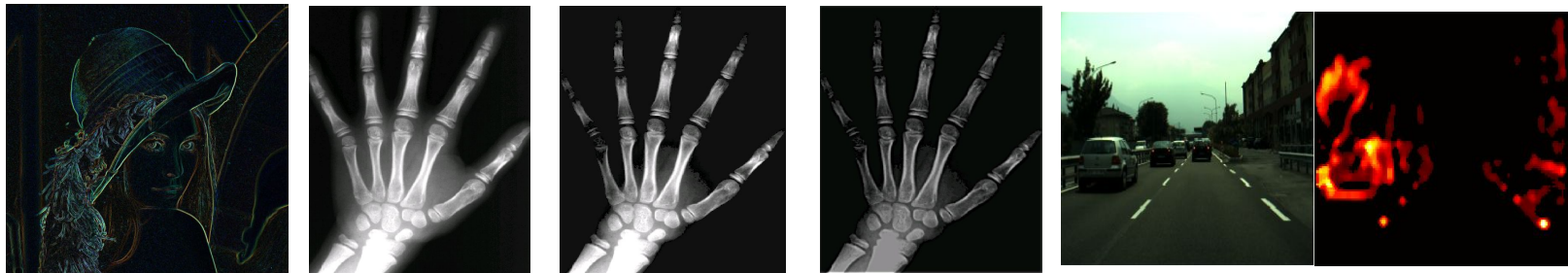


# Levels of Complexity

## Low-Level Vision (Early Vision)

More or less local operations which generate “maps”, that is, point-by-point distributions of values of various quantities.

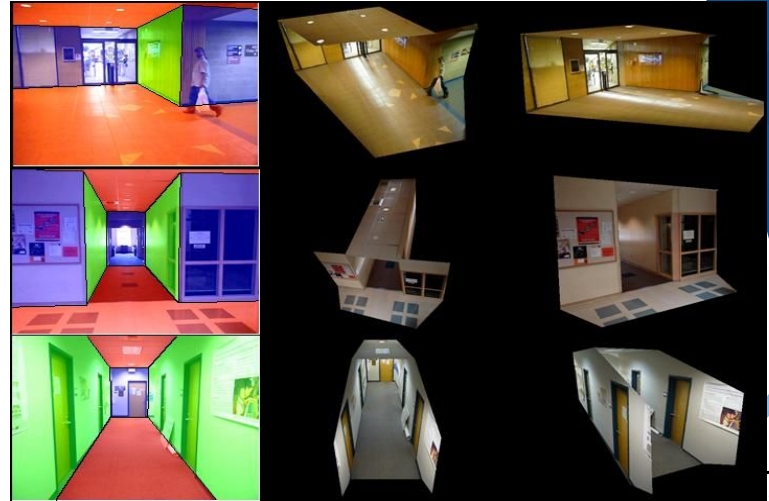
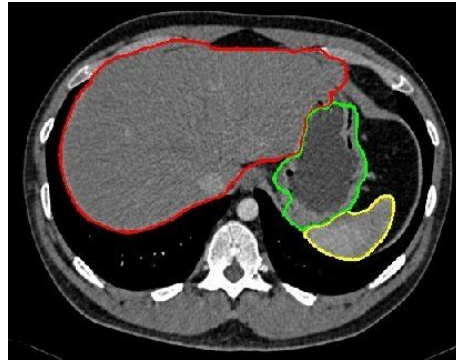
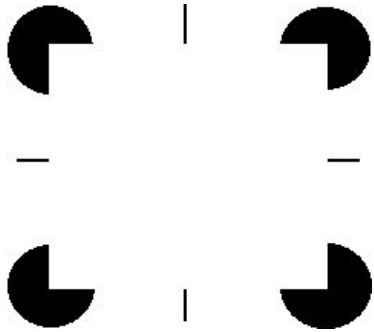
maps of local edge elements, fields of local motion or displacement vectors, depth maps, color constancy;



# Levels of Complexity

## Intermediate-Level Vision

Assembly of the local information from different parts of the image, or from different operations carried out on the same location in the image. For example, image segmentation and contour completion, integration of depth cues. The goal is the reconstruction of visible surfaces and their characteristics.





# Levels of Complexity

## High-Level Vision

- Scene analysis:** Recognition of the objects in a scene. Simple control of behaviors such as steering and obstacle avoidance.
- Active vision:** Making optimal use of the available resources (such as image resolution or computing power) by means of eye movements and exploratory behavior. The simplest case of active vision is the adjustment of the sensor to depth and movement of a viewed object by accommodation, convergence of the eyes, and pursuit eye movements.
- Goal-directed vision:** Control of behaviors based on visual information. High level of recognition, functionality analysis despite large variations in appearance and occlusions.



# Computer Vision vs Image Processing



## Image Processing

Manipulate image data to produce  
**another image**



## Computer Vision

Manipulate image data to produce  
**symbolic information**



# Computer Vision vs Image Processing

## Image Processing:

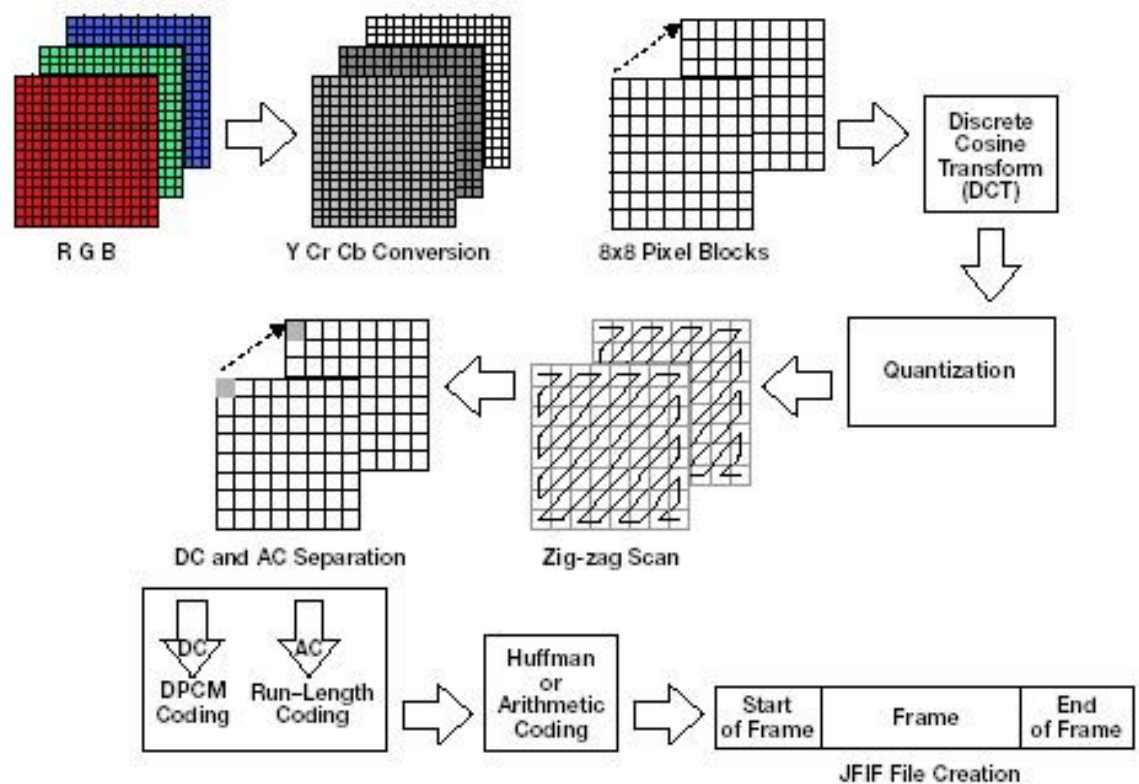
Mostly concerned with image-to-image transformations

- Filtering
- Enhancement
- Compression

# Computer Vision vs Image Processing

## Image Processing:

- *JPEG Compression Algorithm*





# Computer Vision vs Image Processing

## Computer Vision:

Mostly concerned with how the images reflect the world

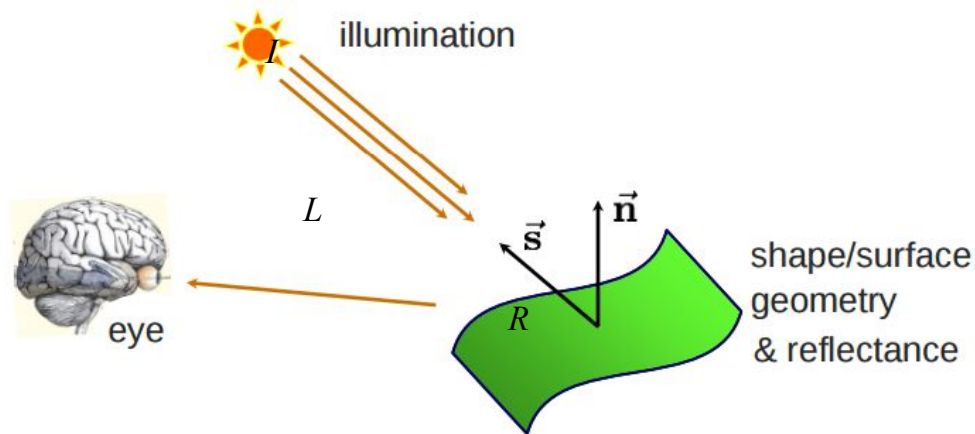
- Filtering Features
- Enhancement for recognition and object detection
- Image Restoration (use of models)
- Compress that preserves some specified features



# Computer Vision Systems

Building blocks of a computer vision system:

## Scene Irradiation



Lambertian  
Model

$$L = R \times H(\vec{n}, \vec{s}, I)$$

Diagram illustrating the Lambertian Model equation:  $L = R \times H(\vec{n}, \vec{s}, I)$ . Red arrows point from descriptive labels to the variables in the equation:

- $L$ : reflected light
- $R$ : surface reflectance (albedo)
- $H$ : incident light
- $\vec{n}$ : surface normal
- $\vec{s}$ : surface direction
- $I$ : light source direction & intensity



# Computer Vision Systems

Building blocks of a computer vision system:

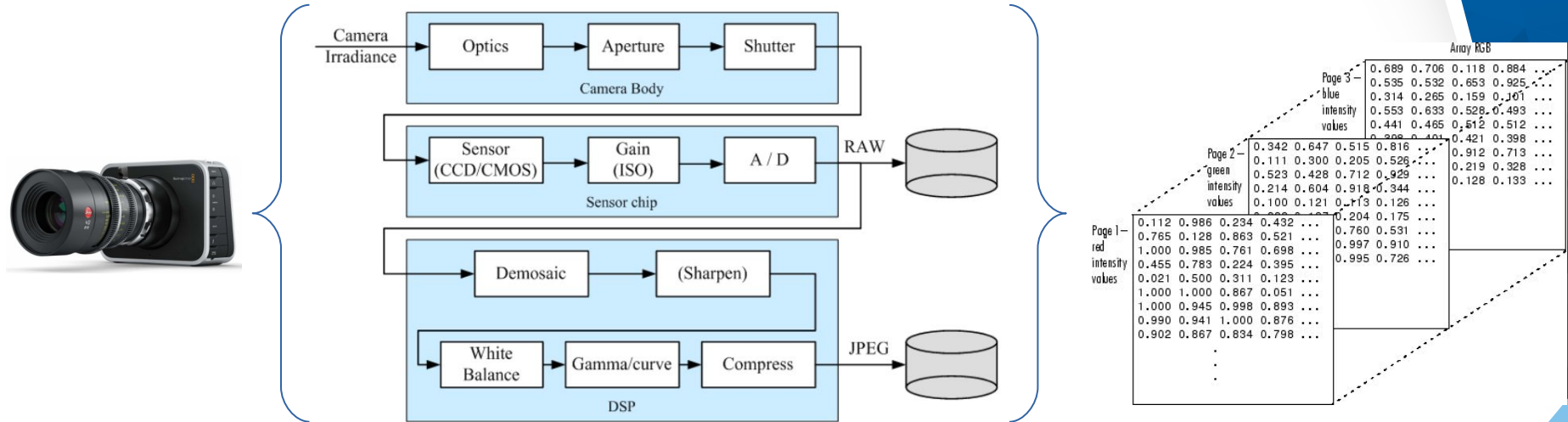
## Hardware



# Computer Vision Systems

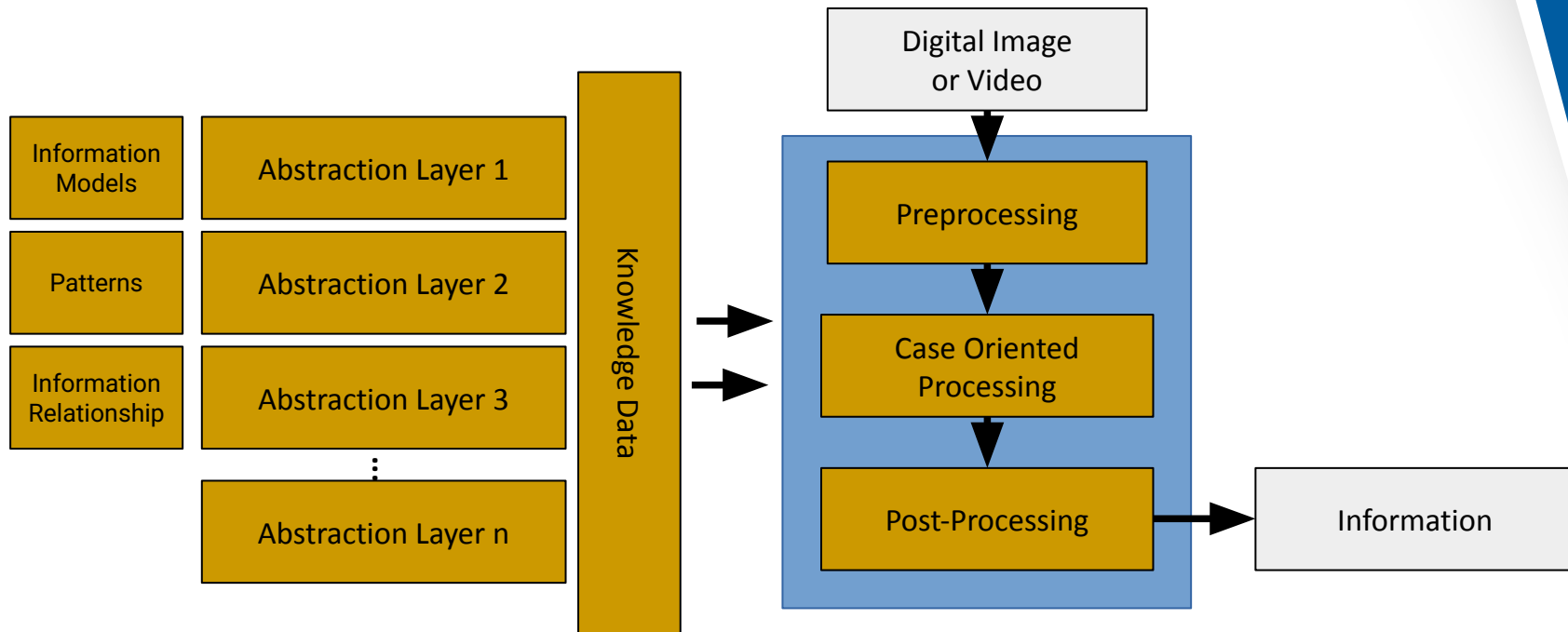
Building blocks of a computer vision system:

Camera → Digital image process



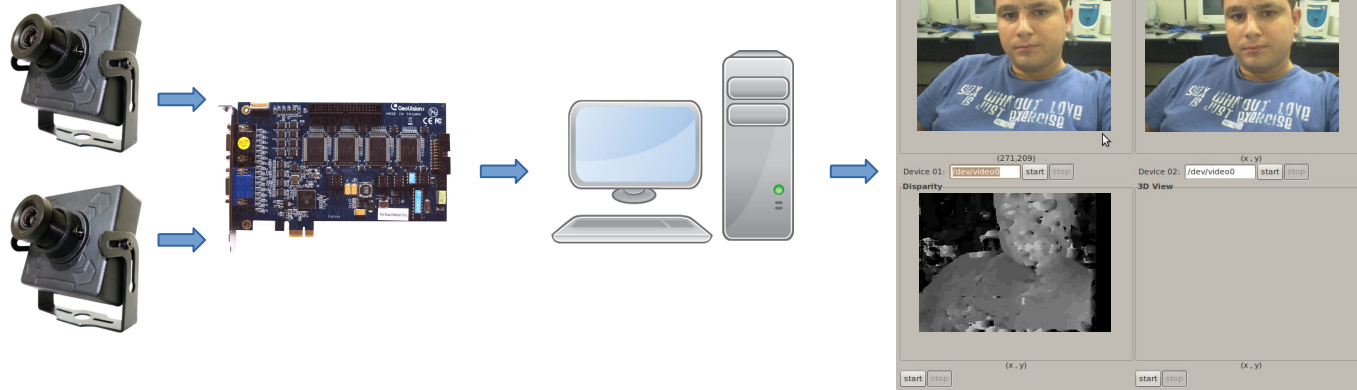
# Computer Vision Systems

Building blocks of a computer vision system:



# Computer Vision Systems

Exemple:



Reconstruction  
Pose Estimation  
Person Recognition  
Volume  
Attitude Classification  
Disease Classification  
Functionality Identification

# Congratulations

**We are at the  
Golden Age of Computer Vision !!**